COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument is a recurring technical step in min-max constructions of embedded minimal hypersurfaces: a cover-refinement on the parameter interval, followed by a bookkeeping argument that turns local near-critical savings into a global supremum improvement. Each author re-derives it in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g) with $H_n(M;\mathbb{Z}) = 0$ (for instance $M = S^{n+1}$), direct minimization of the n-dimensional Hausdorff measure over integral currents yields only the empty hypersurface; a non-trivial minimal hypersurface must be constructed as a saddle-type critical point of the area functional. The construction, due to Almgren and Pitts, is a *min-max* scheme on varifolds; regularity of the output is due to Pitts [Pit81] and Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set $\operatorname{Sing}\Sigma$ has Hausdorff dimension at most $n-7$.*

A key step in Pitts's proof, recurring in [CDL03, DLT13, DLR18, DL16, MN14], is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Theorem 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integer rectifiable varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Theorem 1.2 reduces, following [DLT13, §3.2], to the following claim on the sweepout:[1]

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

The combinatorial core of the proof of the Claim, which is the precise theorem we formalize, reads as follows.

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[1] Throughout this paper, "sweepout" denotes a one-parameter family $\{\Gamma_t\}_{t\in[0,1]}$ in the sense of [DLT13, Def. 3.1]; the multi-parameter min-max setting of [MN14] requires separate combinatorial machinery and lies outside our scope.

Theorem 1.3 (Combinatorial core)**.** *Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open neighborhood $U_t \ni t$ and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exist a finite family of closed pieces $\{p_l \subseteq [0,1]\}_{l\in\iota}$, continuous replacement energies $\{E_l : [0,1] \to \mathbb{R}\}_{l\in\iota}$, and savings $\{s_l \geq 1/(4N)\}_{l\in\iota}$ satisfying:*

(I) *$f - E_l \geq s_l$ on p_l;*
(II) *every $t \in [0,1]$ belongs to at most two of the pieces p_l;*
(III) *$\{t \in [0,1] : f(t) \geq m_0 - 1/N\} \subseteq \bigcup_l p_l$.*

Remark 1.4. The non-trivial content of Theorem 1.3 is the *construction* of the cover $\{p_l\}$ from unstructured witness data: a Lebesgue-number argument on the compact near-critical set produces an initial cover satisfying the skip-2 spacing condition (a) of [DLT13]; a bounded smallest-index induction refines it to an injective piece-to-interval assignment; and a parity argument on the spacing condition delivers the two-fold overlap (II). The bookkeeping corollary below (the existence of a competitor f' with reduced supremum) is a three-line arithmetic consequence.

Translation from the De Lellis–Tasnady setting to Theorem 1.3: the boundary $\partial\Omega_t$ becomes a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U'_t) collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; and the almost-minimizing hypothesis, which in the original proof is the input to Lemma 3.1 of [DLT13], is absorbed into the existence of the local witness (the caller must supply witnesses wherever they would be produced in the ambient GMT setting). The output of Theorem 1.3 is the combinatorially assembled cover data $\{(p_l, E_l, s_l)\}$; the modified sweepout in the original proof is the geometric lift of the scalar competitor produced by the corollary below.

Corollary 1.5 (Sup reduction)**.** *Under the hypotheses of Theorem 1.3, with the pieces $\{p_l\}$ produced there, there exists $f' : [0,1] \to \mathbb{R}$ satisfying:*

(a) *$f'(t) \leq f(t)$ for every $t \in [0,1]$;*
(b) *$f'(t) = f(t)$ whenever $t \notin \bigcup_l p_l$;*
(c) *$\sup f' \leq m_0 - 1/(4N)$.*

Remark 1.6 (Why (a)+(b) are stated)**.** The bound (c) alone is trivialized by any constant $f' \equiv c \leq m_0 - 1/(4N)$. Condition (a) ties f' pointwise below f; condition (b) forces f' to coincide with f wherever no piece modifies it. Together they anchor f' to f, so generic non-constant f rules out constant competitors, and (c) becomes a non-trivial quantitative assertion about the specific competitor the proof produces (concretely, $f'(t) = f(t) - \sum_{l:\, t\in p_l} s_l$).

The bookkeeping underlying Corollary 1.5 is strictly more general than the one-parameter statement it certifies: the same arithmetic applies to any compact, nonempty topological space K with continuous $f : K \to \mathbb{R}$ and any scalars (δ, ε) with $0 < \varepsilon \leq \delta$, taking as input any finite cover of the δ-super-level set of f with per-piece savings $\geq \varepsilon$ and at-most-two overlap. The Lean library ships both forms (Corollaries 2.9 and 2.10 in §2.3); Theorem 1.3 is the combinatorial construction that turns local-witness data into such a cover in the 1D case.

Both results reappear in §2: Theorem 1.3 as the library's main theorem (Theorem 2.7, §2.2), and Corollary 1.5 as the one-parameter sup-reduction corollary (Corollary 2.10, §2.3).

Scope. This note formalizes the proof of the Claim (the combinatorial argument on $[0,1]$) as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships one main theorem (the combinatorial core, §2.2) on $[0,1]$, together with two sup-reduction corollaries (§2.3): a generic compact-K form and its one-parameter

specialization. The formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space) and `f : K` $\to \mathbb{R}$ a continuous scalar energy.

In the proof of Theorem 1.2, the Lemma 3.1 of [DLT13] supplies, for each near-critical parameter t, a local replacement on an $\mathcal{H}^n$-disjoint pair of open sets in M^{n+1} whose boundary energy is reduced by at least $1/(4N)$ on a neighborhood of t. The replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ at a near-critical t has its Hausdorff measure decreased by at least $1/(4N)$ inside U'_t. We keep only the scalar *replacement energy* $E : K \to \mathbb{R}$ and the uniform saving inequality; the geometric family is discarded.

Definition 2.1 (`LocalWitness` structure). [✓ `LocalWitness`]

A `LocalWitness K f t` ε carries an open neighborhood $U \ni t$ in K, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

The proof of the Claim works on the set $K \subset [0, 1]$ of parameters at which the sweepout is within $1/N$ of the min-max value. We use the same convention.

Definition 2.2 (Near-critical set). [✓ `nearCritical`]

For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

At the end of the refinement the data is a family $\{\overline{J_l}\}$ covering K, with each J_l carrying a replacement whose energy saving is at least $1/(4N)$ and with two-fold overlap. We abstract this output: the near-critical threshold $1/N$ becomes a parameter δ and the saving floor $1/(4N)$ a parameter $\varepsilon \leq \delta$.

Definition 2.3 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]

A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l\in\iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

The skip-2 spacing condition (a) of the original proof is a purely one-dimensional geometric condition on a family of open intervals. We isolate it in its own structure so the disjointness lemmas that depend only on spacing can be proved independently of the witness data.

Definition 2.4 (`SkippedSpacedIntervals` structure). [✓ `SkippedSpacedIntervals`]

A `SkippedSpacedIntervals` records a natural number n, centers $c_1, \ldots, c_n \in \mathbf{I}$, positive radii $r_1, \ldots, r_n > 0$, and the skip-2 spacing axiom $c_i + r_i < c_{i+2} - r_{i+2}$ whenever $i + 2 \leq n$. The i-th open interval $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ is an abbreviation on this structure.

On top of the geometric structure, the initial cover also carries a witness center $w_i \in$ NC with an attached local witness for each interval, together with the containment of each I_i inside the corresponding witness neighborhood. In our formalization the interval center c_i and the witness center w_i are recorded separately: grid points need not themselves lie in NC.

Definition 2.5 (`InitialCover` structure). [✓ `InitialCover`]

Specialized to $K = \mathbf{I}$. An `InitialCover X f m₀ N` carries the data of a `SkippedSpacedIntervals` (Definition 2.4, satisfying spacing condition (a)), plus, for each $i = 1, \ldots, n$, a witness center

$w_i \in \mathrm{NC}$ and a local witness at w_i with saving $1/(4N)$, subject to (b) containment: $I_i \subseteq U_{w_i}$; (c) coverage: $\mathrm{NC} \subseteq \bigcup_i I_i$.

The refinement is described as an iteration: each I_j with $j \geq 2$ is either kept whole or split into two halves. The state of this iteration after ℓ moves is the following.

Definition 2.6 (`PartialRefinement` structure). [✓ `PartialRefinement`]

Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. **Public API: combinatorial core.** The library's main theorem is the existence of a refined multiplicity-bounded cover of the near-critical set carrying the witness savings: the non-trivial combinatorial content extracted from the original proof. Its hypotheses coincide with the intro statement (Theorem 1.3); the output is a `FiniteCoverWithWitnesses` (Definition 2.3).

Theorem 2.7 (Combinatorial core). [✓ `exists_refinement`]

Let $f : \mathbf{I} \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose every $t \in \mathrm{NC}$ admits a `LocalWitness` at t with saving $1/(4N)$ (Definition 2.1). Then there exists a `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ (Definition 2.3): a finite family of closed pieces $\{p_l \subseteq \mathbf{I}\}$, continuous replacement energies $\{E_l : \mathbf{I} \to \mathbb{R}\}$, and savings $\{s_l \geq 1/(4N)\}$ satisfying conditions (I)–(IV).

The proof is given in §3 as three chained stages: Stage A builds an initial cover from a Lebesgue-number argument on NC; Stage B runs a bounded smallest-index induction to a terminal partial refinement; Stage C assembles the output by extending the open-neighborhood saving bound to the closure via continuity and deriving two-fold overlap via the parity-gap disjointness argument.

2.3. **Public API: sup-reduction corollaries.** The arithmetic that turns a `FiniteCoverWithWitnesses` into a competitor with reduced supremum is a short scalar lemma independent of covers, pieces, or multiplicity: any competitor that saves a fixed amount on the relevant super-level set yields the same amount of supremum reduction.

Lemma 2.8 (Scalar sup reduction). [✓ `sup_le_of_saving`]

Let K be a nonempty set and $f : K \to \mathbb{R}$ a bounded function with $m_0 = \sup f$. Let $g : K \to \mathbb{R}$ satisfy

(i) *$g(t) \leq f(t)$ for every $t \in K$;*
(ii) *$f(t) - g(t) \geq \varepsilon$ whenever $f(t) \geq m_0 - \delta$,*

where $0 < \varepsilon \leq \delta$. Then $\sup g \leq m_0 - \varepsilon$.

Proof. Fix $t \in K$. If $f(t) \geq m_0 - \delta$, then (ii) gives $g(t) \leq f(t) - \varepsilon \leq m_0 - \varepsilon$. If $f(t) < m_0 - \delta$, then (i) gives $g(t) \leq f(t) < m_0 - \delta \leq m_0 - \varepsilon$. Taking the supremum over t completes the proof. □

Lemma 2.8 is a general statement about lowering the supremum of a bounded real-valued function by exhibiting a competitor with a pointwise saving on the super-level set $\{f \geq m_0 - \delta\}$; no sweepout, cover, or multiplicity hypothesis enters. It applies verbatim whenever a pointwise saving can be established on the "where it matters" super-level set. The Lean library ships two corollaries of Theorem 2.7 via this lemma: the generic-K form (any `FiniteCoverWithWitnesses`), and the one-parameter specialization to $\mathbf{I}$ that chains through Theorem 2.7.

Corollary 2.9 (Sup reduction, generic form). [✓ `exists_sup_reduction_of_cover`]

Let K be a compact, nonempty topological space, $f : K \to \mathbb{R}$ continuous with $m_0 = \sup f$, and $0 < \varepsilon \leq \delta$. Given a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ (Definition 2.3) with pieces $\{p_l\}$, there exists $f' : K \to \mathbb{R}$ satisfying:

(a) *$f'(t) \leq f(t)$ for every $t \in K$;*

(b) $f'(t) = f(t)$ *whenever* $t \notin \bigcup_l p_l$*;*
(c) $\sup f' \le m_0 - \varepsilon$*.*

Proof. Set $f'(t) := f(t) - \sum_{l\,:\,t\in p_l} s_l$. Since each $s_l > 0$, (a) holds. If $t \notin \bigcup_l p_l$ the sum is empty, giving (b). For (c), apply Lemma 2.8: (i) follows from (a); (ii) holds because when $f(t) \ge m_0 - \delta$, (IV) gives some $p_l \ni t$, (III) gives $s_l \ge \varepsilon$, and the remaining at-most-one saving (by two-fold overlap (II)) is non-negative, so $f(t) - f'(t) \ge \varepsilon$. The lemma yields $\sup f' \le m_0 - \varepsilon$. □

Corollary 2.10 (One-parameter sup reduction)**.** [✓ `exists_sup_reduction`]
Under the hypotheses of Theorem 2.7, there exist $f' : \mathbf{I} \to \mathbb{R}$ *and a set* $S \subseteq \mathbf{I}$ *containing* NC *such that:*

(a) $f'(t) \le f(t)$ *for every* $t \in \mathbf{I}$*;*
(b) $f'(t) = f(t)$ *whenever* $t \notin S$*;*
(c) $\sup f' \le m_0 - 1/(4N)$*.*

Corollary 2.10 is Corollary 2.9 applied to the `FiniteCoverWithWitnesses` produced by Theorem 2.7, with $(\delta, \varepsilon) = (1/N, 1/(4N))$; the modification set S is the union of the cover pieces, which contains NC by condition (IV) of Definition 2.3.

2.4. **Internal lemmas.** The declarations below are the sub-lemmas of Theorem 2.7; proofs are sketched in §3.

Proposition 2.11 (Compactness of NC)**.** [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]
If K *is compact and* f *continuous,* NC *is compact. If additionally* $m_0 = \sup f$ *and* $N > 0$*,* NC *is nonempty.*

- ✓ `exists_initialCover` : existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` : existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` : at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` : the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.

The assembly itself—packaging the Stage A–B–C output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$—is Theorem 2.7 itself, whose Lean declaration ✓ `exists_refinement` chains the sub-lemmas above.

Figure 1 shows the layered architecture.

3. Proof structure

We split the proof of Theorem 2.7 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. The sup-reduction corollaries (Corollary 2.9 and Corollary 2.10) follow by Lemma 2.8 applied to the resulting cover; we sketch that 3-line argument after Stage C.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover)**.** [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.7, an initial cover (Definition 2.5) exists.

Proof. Compactness of NC (Proposition 2.11) plus the open cover $\{U_t\}_{t\in\mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \dots, M$, and select those c_k within $\lambda/4$ of NC.

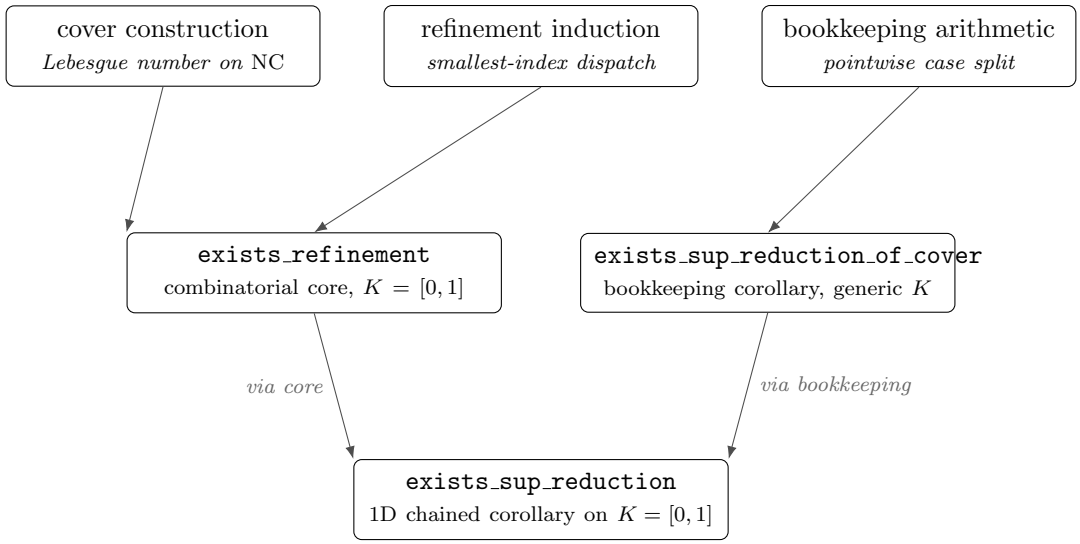

FIGURE 1. `CombArg` v0.2 conceptual architecture. Three combinatorial pillars (cover construction via Lebesgue number, refinement induction via smallest-index dispatch, bookkeeping arithmetic via pointwise case split) feed two main theorems: the combinatorial core `exists_refinement`, which produces a finite cover with two-fold overlap on $K = [0, 1]$, and the bookkeeping corollary `exists_sup_reduction_of_cover`, which turns any such cover into a sup-reducing competitor on a generic compact K. The chained 1D corollary `exists_sup_reduction` composes the two on $K = [0, 1]$ and is the user-facing entry point.

For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. **Stage B: the refinement induction.** Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case)**.** [✓ `step_zero`]

step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3 (Inductive step)**.** [✓ `step_succ_at`]

Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \ldots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.

(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell + 1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

Proposition 3.4 (Termination)**.** [✓ `exists_terminal_refinement`]

Starting from step_zero(ic) *and iterating* step_succ_at*, one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

[2]A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

Proof. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr′ with $i^* \notin$ rem(pr′) and rem(pr′) $\subseteq$ rem(pr)$\setminus\{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell+1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. **Stage C: assembly.** Stage C packages the refined covering as a `FiniteCoverWithWitnesses` (Definition 2.3), which the core theorem turns into the sup-reducing f'. We trace through [DLT13, §3.2, Step 2] in sequence, indicating at each step where the content lands in the formalization.

De Lellis and Tasnady begin by applying Lemma 3.1 to produce, at each near-critical parameter t, a replacement family $\{\partial\Omega_{i,t}\}$ with three properties: $\Omega_{i,t} = \Omega_t$ outside (a_i, b_i) and agrees with Ω_t outside U_i' inside (a_i, b_i); the global upper bound $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) + \frac{1}{4N}$; and, most importantly, the saving $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) - \frac{1}{2N}$ when $t \in (a_i + \delta, b_i - \delta)$. In our formalization this data is what a `LocalWitness` (Definition 2.1) carries: the scalar $E_{\sigma(k)} : K \to \mathbb{R}$ plays the role of $\mathcal{H}^n(\partial\Omega_{i,t})$, the neighborhood U_t plays (a_i, b_i), and the two separate bounds ("saving $-\frac{1}{2N}$ on the inner core" and "no-loss $+\frac{1}{4N}$ outside") collapse to a single uniform saving $\varepsilon = \frac{1}{4N}$ on U_t.[3] Stage B has already handed us pieces p_l on which this witness saving is valid (Proposition 3.6), so by Stage C we have per-piece scalar savings $s_l \ge \varepsilon$ directly.

The next observation in the proof is that if $t \in (a_i, b_i) \cap (a_j, b_j)$, then $j = i + 1$, and moreover $\operatorname{dist}(U_i', U_{i+1}') > 0$. This is the two-fold overlap of the refined covering, and in our formalization it is the `twoFold` field of `FiniteCoverWithWitnesses`, derived from the parity-gap spacing by Proposition 3.5. The parity-gap lemmas themselves live on the abstract `SkippedSpacedIntervals` (Definition 2.4) and are therefore independent of the witness data: `InitialCover` provides access to them by projecting out the underlying geometry.

The modified sweepout $\{\partial\Omega_t'\}$ is then defined piecewise: $\Omega_t' = \Omega_t$ outside all J_i; $\Omega_t' = \Omega_{i,t}$ if t lies in a single J_i; and on $J_i \cap J_{i+1}$ the sweepout blends the two replacements, $\Omega_t' = [\Omega_t \setminus (U_i' \cup U_{i+1}')] \cup [\Omega_{i,t} \cap U_i'] \cup [\Omega_{i+1,t} \cap U_{i+1}']$. Stripped of the geometry this is additive scalar subtraction of savings over the pieces containing t: our reduced energy is

$$f'(t) \;=\; f(t) \;-\; \sum_{l\,:\,t\in\overline{J_l}} s_l,$$

and the two-fold invariant bounds the sum length by 2, matching the two-piece blend on overlaps.

The area accounting then splits on whether t is near-critical. For $t \notin K$, the point lies in at most two J_i's and each replacement can add at most $\frac{1}{4N}$, so

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) + \tfrac{1}{2N}.$$

For $t \in K$, the point lies in at least one inner core $(a_i + \delta, b_i - \delta)$, where the replacement saves $\ge \frac{1}{2N}$ in U_i', and in at most one other J_l where it costs $\le \frac{1}{4N}$ in U_l', for net

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) - \tfrac{1}{4N}.$$

In our formalization the same case analysis is the scalar argument for $f'(t) \le m_0 - \varepsilon$ inside the abstract core (the lemma `reducedEnergy_le` from §2.2): split on whether t lies in any piece of the cover, use the saving floor $s_l \ge \varepsilon$ in the positive branch and the covering condition $\{f \ge m_0 - \delta\} \subseteq \bigcup_l p_l$ in the negative branch. With $(\delta, \varepsilon) = (1/N, 1/(4N))$ this specializes to the arithmetic of the original proof; the abstract core reads off the same algebra for arbitrary (δ, ε) with $\varepsilon \le \delta$.

[3] This is strictly stronger than the original two-bound statement: the local witness asserts a uniform saving of $\frac{1}{4N}$ on all of U_t, whereas De Lellis–Tasnady only require a larger saving $\frac{1}{2N}$ on the inner core $(a_i + \delta, b_i - \delta)$ and bounded loss $\frac{1}{4N}$ elsewhere. The stronger abstract assumption removes the δ parameter of the inner core from the scalar theorem; the caller must verify the uniform version holds in the ambient setting.

The rest of this subsection establishes the two propositions that Stage B leaves open: the saving bound extends from open J_k to $\overline{J_k}$ (continuity), and the two-fold invariant holds (a parity argument on the spacing condition (a)).

Proposition 3.5 (TwoFold via parity)**.** [✓ `terminal_twoFold`]

For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \ldots, \overline{J_L}$.*

Proof. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i-j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \ldots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \ldots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c-a$ is even (and necessarily ≥ 2), or $c-a$ is odd and ≥ 3, in which case $b-a$ and $c-b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound)**.** [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \geq 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Proof. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \geq 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.1 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\geq$-inequalities under limits. □

Proof of Theorem 2.7. Chain Propositions 3.1 and 3.4: the former produces an `InitialCover` from the witness hypothesis, and the latter yields a terminal `PartialRefinement` pr of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i. Assemble the `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ by taking pieces $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and E_k the replacement energy of the $\sigma(k)$-th local witness. Invariant (I) is Proposition 3.6; invariant (II) is Proposition 3.5; invariant (III) holds by construction ($s_k = 1/(4N)$); coverage of NC chains as $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Corollary 2.9 then follows by the three-line bookkeeping proved in §2.3 applied to this `FiniteCoverWithWitnesses`; Corollary 2.10 is the specialization to $(\delta, \varepsilon) = (1/N, 1/(4N))$.

4. Use as a software dependency

Suppose the min-max setting from the literature has been formalized in Lean: an admissible class $\mathcal{A}$ of sweepouts, the inf-sup characterization $m_0 = \inf_{\Omega \in \mathcal{A}} \sup f_\Omega$, the local replacement construction (De Lellis–Tasnady's Lemma 3.1 [DLT13]), and the Hausdorff measure of boundary surfaces. This section describes how the abstract declarations of `CombArg` then *lift into the combinatorial step of the original proof* from the literature.

The library is the formal abstract version of the combinatorial slice that [Pit81, CDL03, DLT13, DLR18, MN14] each re-derive in their own notation. Its four public declarations make no GMT references and stand for the four pieces the literature recombines at every instantiation. `LocalWitness` (K, f, t, ε) (Definition 2.1) carries the abstract per-parameter local-reducer data; ✓ `exists_refinement` is the formal counterpart of [DLT13] §3.2 Step 1, the interval refinement that turns a family of local witnesses on the near-critical set into a finite cover with two-fold overlap; ✓ `exists_sup_reduction_of_cover` is the formal counterpart of [DLT13] §3.2 Step 2, the scalar bookkeeping that turns such a cover into a sup-reducing competitor; and ✓ `exists_sup_reduction` composes these two into a single one-parameter call.

What the literature supplies in its prose, and what the min-max setting must formalize on its side, is the data that turns these abstract declarations into concrete proof artifacts. Concretely:

the continuous boundary energy $f := \Omega.\texttt{boundaryEnergy} : \mathbf{I} \to \mathbb{R}$ of a sweepout Ω realizing $m_0 = \sup f$; a near-criticality parameter $N \in \mathbb{N}$ chosen so that $1/(4N)$ beats whatever contradiction window the admissible class needs; and a local replacement lemma producing, at every parameter t with $f(t) \geq m_0 - 1/N$, a `LocalWitness` $(\mathbf{I}, f, t, 1/(4N))$ whose data fields come from the GMT-side replacement family. The library's Lean declarations do not change between the abstract and instantiated forms — they are exactly the same definitions either way; what changes is that this min-max-side data flows in as the inputs.

4.1. **The instantiation: `LocalWitness` from the local replacement lemma.** The `LocalWitness` structure was designed to be the data shape that the De Lellis–Tasnady local replacement lemma [DLT13, Lemma 3.1] produces, with each field naming a specific piece of the GMT-side replacement data. The open interval $(a_i, b_i) \subseteq [0, 1]$ on which the local replacement saves energy becomes the `neighborhood` field; the boundary energy of the replaced sweepout, $s \mapsto \mathcal{H}^n(\partial\widetilde{\Omega}_s)$ with $\widetilde{\Omega}_s$ the original sweepout with the local replacement inserted at parameter s, becomes `replacementEnergy`; its continuity in s (which [DLT13] treats implicitly through the continuity of the replaced family in the inserted parameter) becomes the `replacementEnergy_continuous` obligation that the GMT side must discharge explicitly; and the quantitative inequality $f(s) - \mathcal{H}^n(\partial\widetilde{\Omega}_s) \geq 1/(4N)$ on (a_i, b_i) becomes `saving_bound`.

4.2. **The proof chain: from sup reduction to contradiction.** With this data in place, ✓ `exists_sup_reduction` returns a scalar competitor $f' : \mathbf{I} \to \mathbb{R}$ together with a modification set $S \subseteq \mathbf{I}$ satisfying $\{t \in \mathbf{I} : f(t) \geq m_0 - 1/N\} \subseteq S$ (coverage), the pointwise dominance $f'(t) \leq f(t)$ everywhere (a), the localization $f'(t) = f(t)$ for $t \notin S$ (b), and the sup reduction $\sup_t f'(t) \leq m_0 - 1/(4N)$ (c). This is — definitionally — the sup-reduction step the original proof uses. The competitor f' and modification set S are the formal images of *the sweepout's energy after the local replacements have been combined*; conditions (a) and (b) anchor f' to f so that f' is a genuine combinatorial reduction of f rather than a trivial constant.

The literature now closes the proof by lifting f' back to a sweepout $\Omega' \in \mathcal{A}$ and contradicting the inf-sup. In Lean this chain has three pieces, of which the library discharges the first: ✓ `exists_sup_reduction` produces f' with $\sup f' \leq m_0 - 1/(4N)$. The min-max setting then discharges the second by taking f' together with its modification set S back to a sweepout $\Omega' \in \mathcal{A}$ — the (b) localization guarantee makes this lift mechanical, since Ω' agrees with Ω outside S and inserts the local replacement on each piece of S. The contradiction follows immediately from $\sup f_{\Omega'} \leq m_0 - 1/(4N) < m_0 = \inf_{\mathcal{A}} \sup$ together with $\Omega' \in \mathcal{A}$. Together with the library's first piece, these reconstruct line for line the literature's contradiction chain.

4.3. **Skeleton client code.** The pattern below shows how a downstream GMT formalization would invoke `CombArg`; it adapts `examples/MinimalUsage.lean` in the repository, with `YourGMT.*` stubs standing in for the GMT-side declarations a consumer formalization would supply:

```
import CombArg
import YourGMT  -- consumer's formalization

theorem minmax_contradiction
    (O : YourGMT.SweepoutType) (hO : O in YourGMT.admissibleClass)
    (h_minmax : O.boundaryEnergy.sup = YourGMT.infMinmax) :
    False := by
  set f := O.boundaryEnergy
  set m0 := sSup (Set.range f)
  obtain <N, hN_pos, hN_window> := YourGMT.choose_N O
  -- Lean image of DLT Lemma 3.1: per-parameter local witness.
  have witness : forall t : unitInterval, f t >= m0 - 1 / (N : R) ->
```

```
                LocalWitness unitInterval f t (1 / (4 * (N : R))) :=
  fun t ht => YourGMT.localWitness_of_DLT O t ht hN_pos
-- comb-arg: combinatorial bookkeeping -> scalar sup reduction.
obtain <f', S, _, h_le, h_eq, h_sup> :=
  CombArg.exists_sup_reduction
    O.boundaryEnergyContinuous rfl hN_pos witness
-- Lift f' back to a sweepout in admissibleClass.
obtain <O', hO', hO'_energy> :=
  YourGMT.lift_sweepout f' h_le h_eq O hO
-- Contradiction.
have h_lt : O'.boundaryEnergy.sup < m0 := by
  rw [hO'_energy]; linarith [h_sup, hN_window]
exact absurd h_lt (YourGMT.le_of_admissible hO' h_minmax)
```

The three `YourGMT.*` identifiers mark the GMT-side responsibility: `YourGMT.choose_N` picks N so that $1/(4N)$ beats the admissible class's contradiction window; `YourGMT.localWitness_of_DLT` is the GMT formalization of the local replacement lemma [DLT13, Lemma 3.1], returning a per-t `LocalWitness`; and `YourGMT.lift_sweepout` converts the scalar f' (together with its modification set S) back into a sweepout in $\mathcal{A}$.

4.4. **Known friction.** The library is currently specialized to the 1D parameter space $K = \mathbf{I}$; multi-parameter sweepouts ($K = [0,1]^m$, in particular Almgren cycles) will need a multi-parameter cover construction, planned for v0.3. The output is scalar rather than geometric: the consumer receives f' with its sup bound and the modification set S, and must lift back to a sweepout in $\mathcal{A}$ on the GMT side, with the (b) off-S agreement designed to make this lift mechanical. Continuity of `replacementEnergy` in the inserted parameter falls on the consumer to discharge explicitly, since the local replacement lemma supplies it only implicitly through the continuity of the replaced family. The library is pinned to `leanprover/lean4:v4.30.0-rc2` together with the Mathlib revision in `lake-manifest.json`, so bumps on either side may require coordination. The library itself uses only the three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`), and standard Mathlib measure-theory work stays within the same three.

References

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